Deploying a Highly Available Global E-commerce Frontend using AWS CloudFront & S3 :

Scenario:
You are tasked with setting up a secure and scalable web application infrastructure on AWS for a fictional e-commerce website called "CloudMart." CloudMart aims to serve customers globally with high availability and low latency. You will utilize AWS services including CloudWatch, CloudFront,S3, to achieve the desired architecture.

Lab Tasks:

Setting up S3 Bucket:

- Create an S3 bucket named "cloudmart-static" to store static website assets such as images, CSS, and JavaScript files.

Configuring CloudFront:

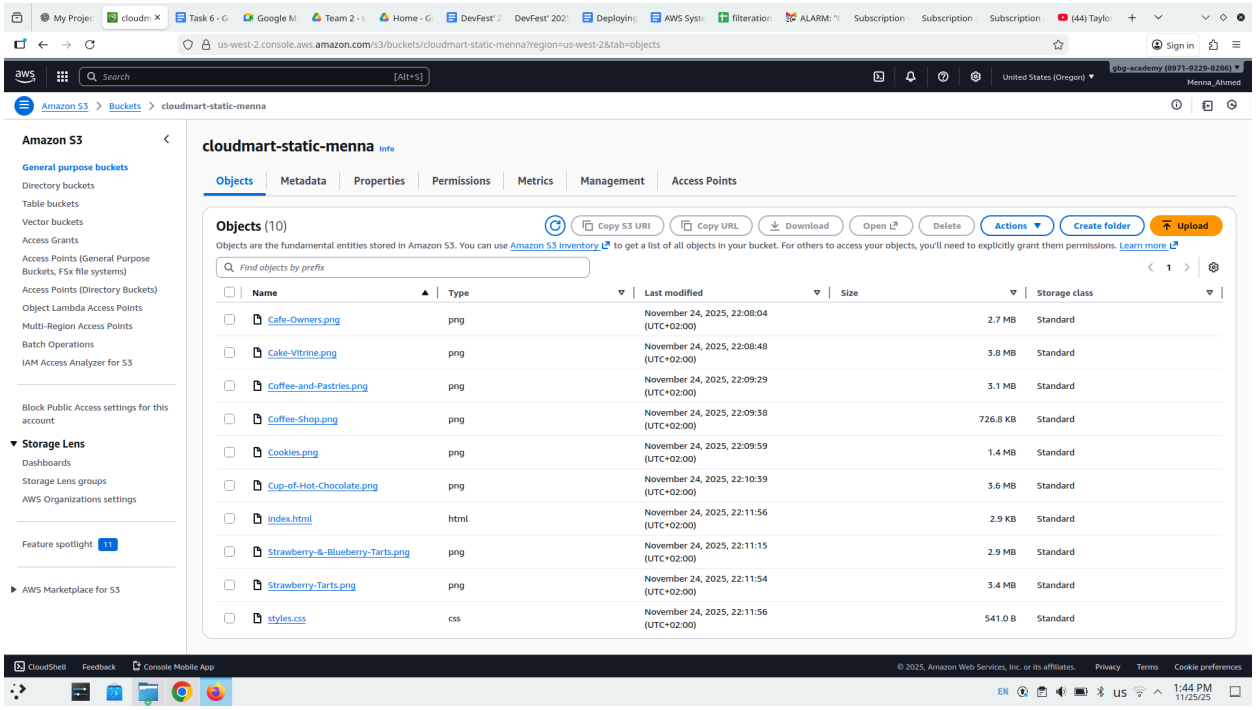
- Set up a CloudFront distribution to serve the static content globally with low latency.
- Configure the CloudFront distribution to use the S3 bucket as its origin.

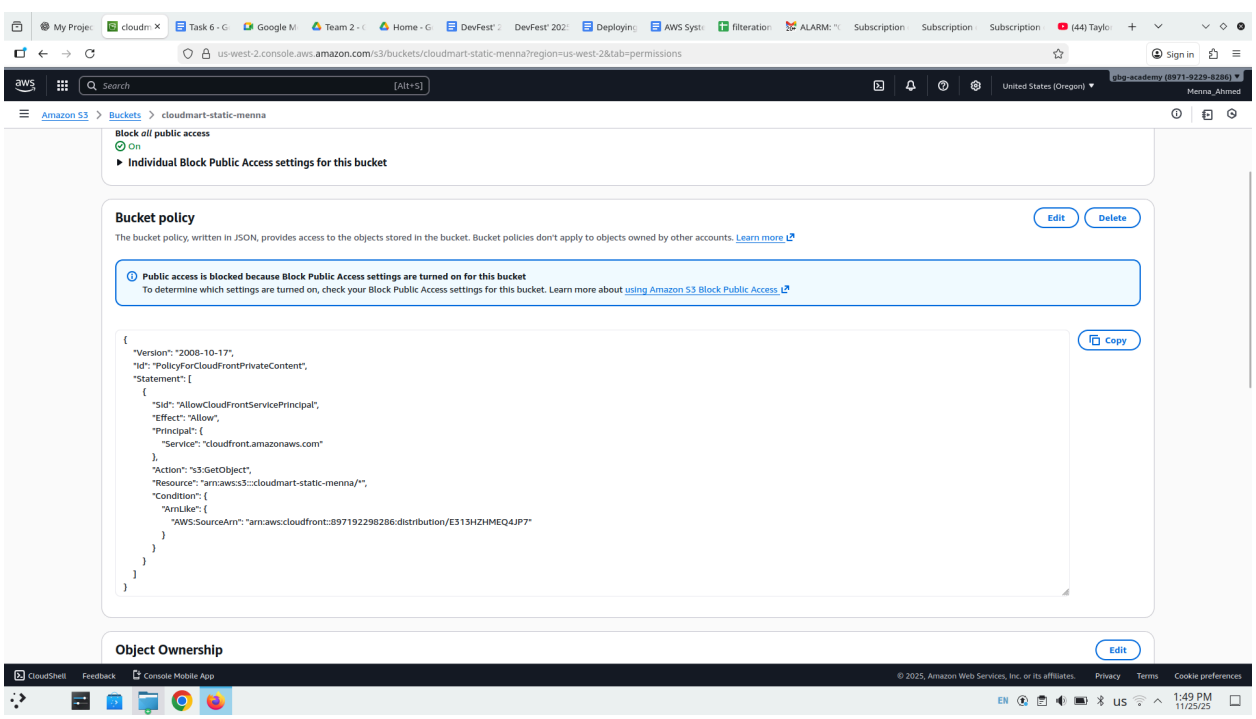
Monitoring with CloudWatch:

- Set up CloudWatch alarms to monitor the health of the infrastructure.
- Create alarms for metrics such as latency, request count, and 4xx/5xx

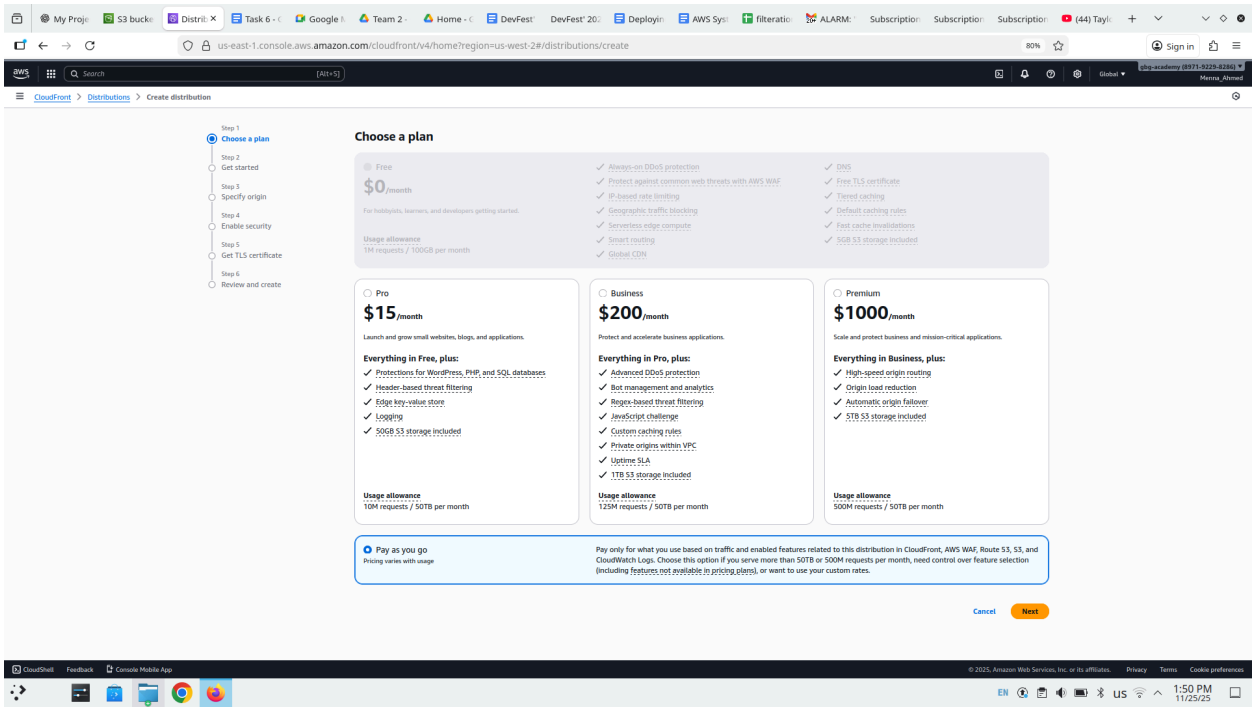
Additional Point: Restricting Direct Access to S3 — CloudFront Only

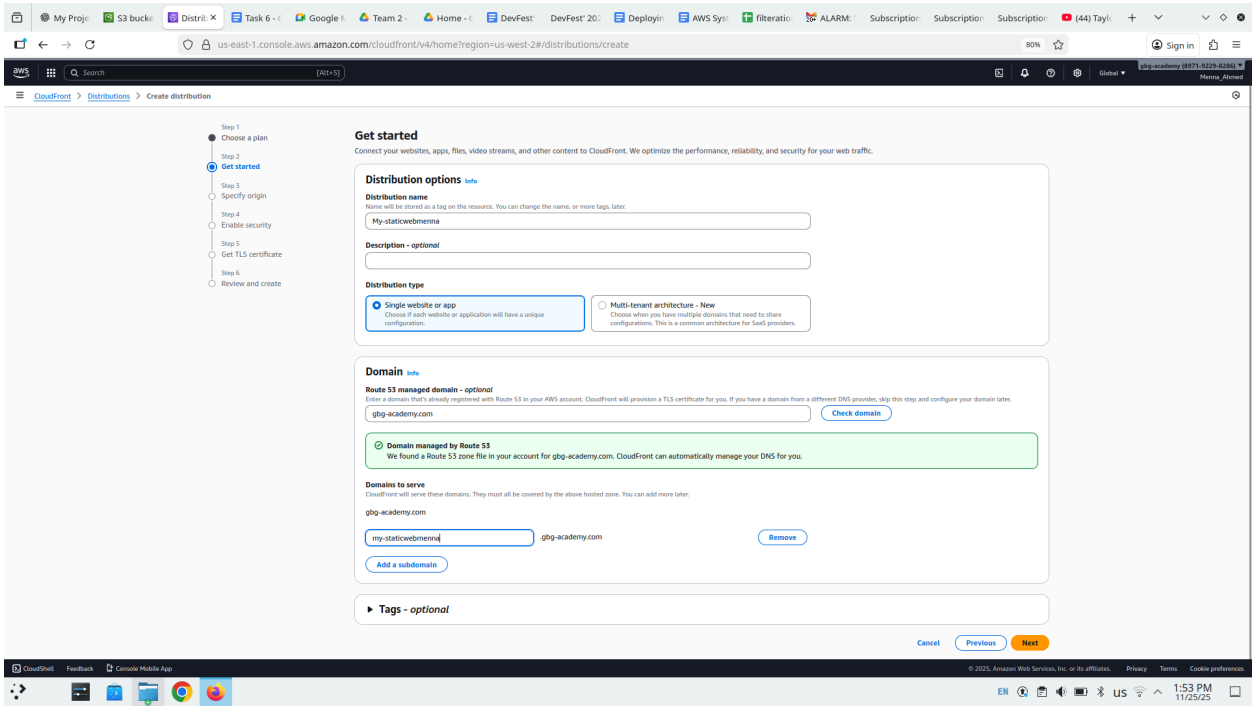
Create S3

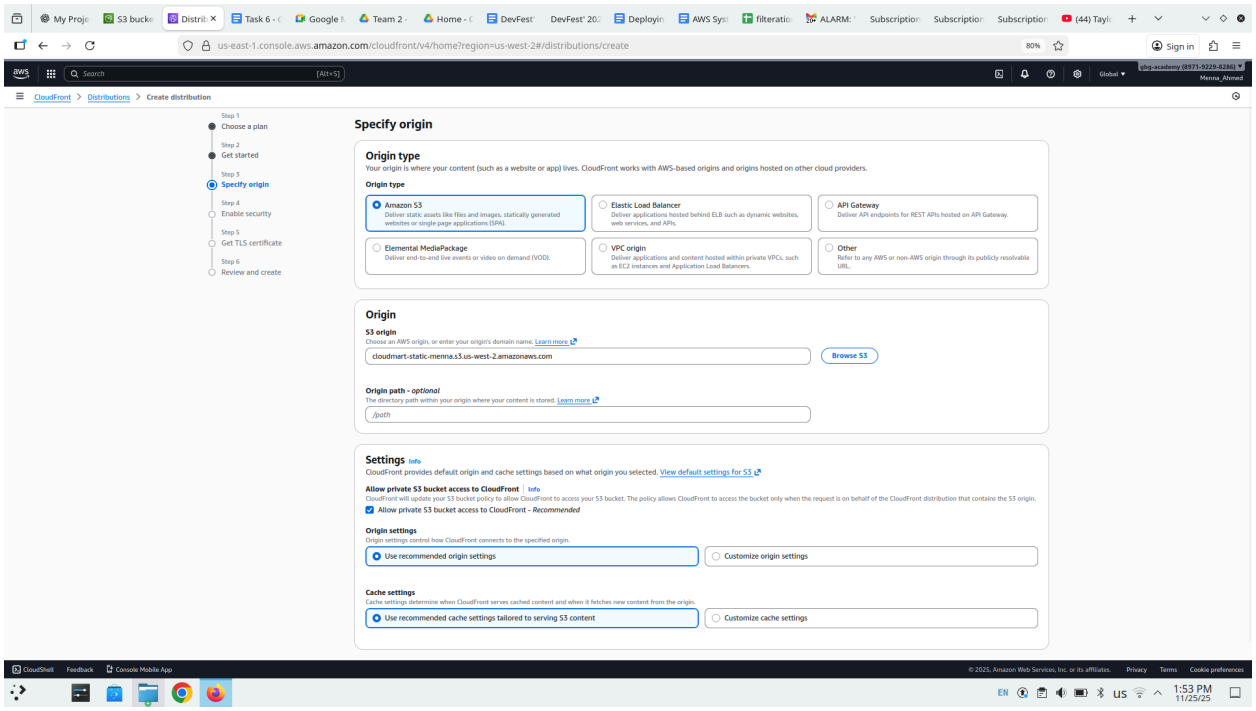

us-west-2.console.aws.amazon.com/s3/buckets/cloudmart-static-menna?region=us-west-2&tab=objects
Amazon S3 > Buckets > cloudmart-static-menna
Amazon S3
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Access Grants
Access Points (General Purpose Buckets, FSx file systems)
Access Points (Directory Buckets)
Object Lambda Access Points
Multi-Region Access Points
Batch Operations
IAM Access Analyzer for S3
Block Public Access settings for this account
Storage Lens
Dashboards
Storage Lens groups
AWS Organizations settings
Feature spotlight 11
AWS Marketplace for S3
cloudmart-static-menna Info
Objects
Metadata
Properties
Permissions
Metrics
Management
Access Points
Objects (10)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
Cafe-Owners.png
Cake-Vitrine.png
Coffee-and-Pastries.png
Coffee-Shop.png
Cookies.png
Cup-of-Hot-Chocolate.png
index.html
Strawberry-&-Blueberry-Tarts.png
Strawberry-Tarts.png
styles.css


us-west-2.console.aws.amazon.com/s3/buckets/cloudmart-static-menna?region=us-west-2&tab=permissions
Amazon S3 > Buckets > cloudmart-static-menna
Block all public access
On
Individual Block Public Access settings for this bucket
Bucket policy
Edit
Delete
The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. Learn more
Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about using Amazon S3 Block Public Access
Copy
{
"Version": "2008-10-17",
"Id": "PolicyForCloudFrontPrivateContent",
"Statement": [
{
"Sid": "AllowCloudFrontServicePrincipal",
"Effect": "Allow",
"Principal": {
"Service": "cloudfront.amazonaws.com"
},
"Action": "s3:GetObject",
"Resource": "arn:aws:s3:::cloudmart-static-menna/*",
"Condition": {
"ArnLike": {
"AWS:SourceArn": "arn:aws:cloudfront::897192298286:distribution/E313HZHMEQ4JP7"
}
}
}
]
}
Object Ownership
Edit

Now create CloudFront Distribution

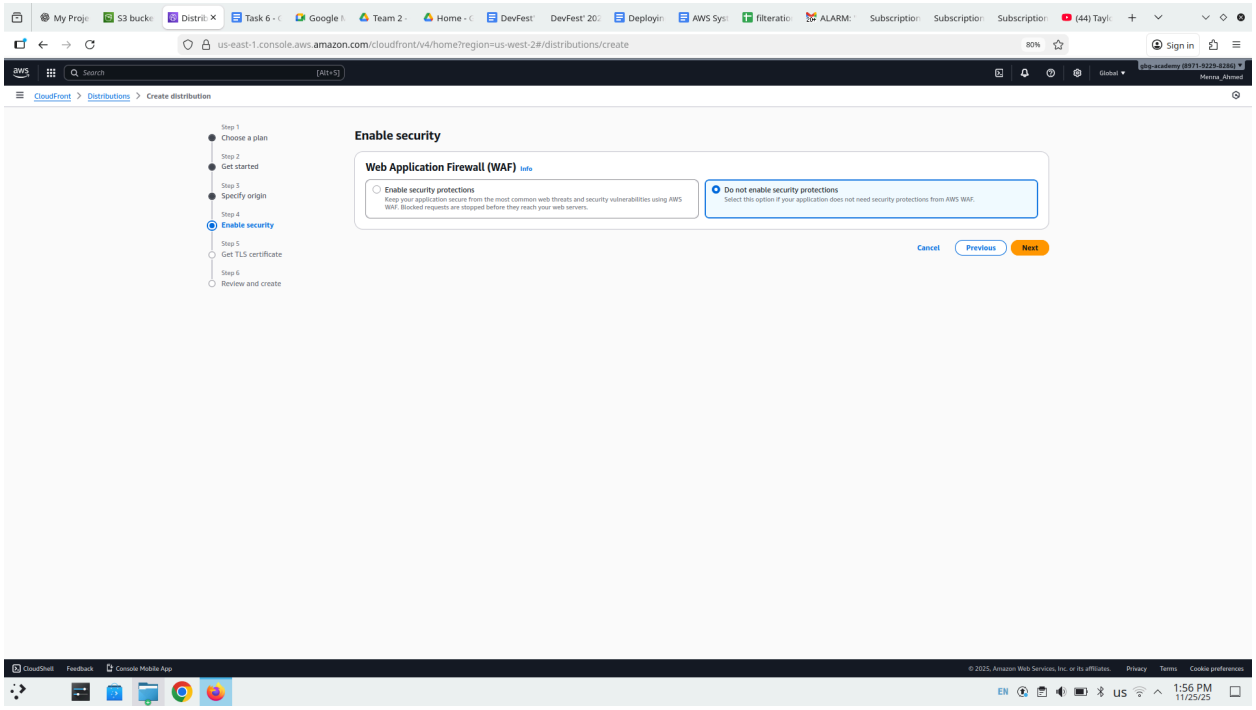




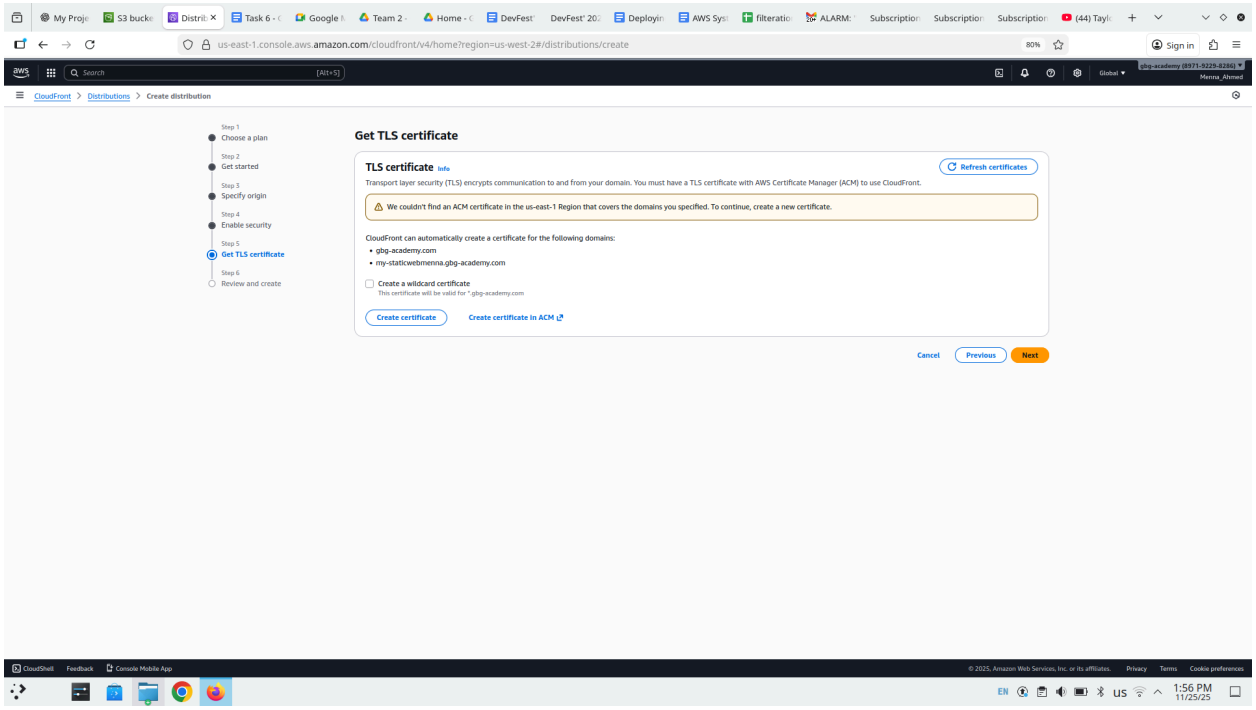


Next

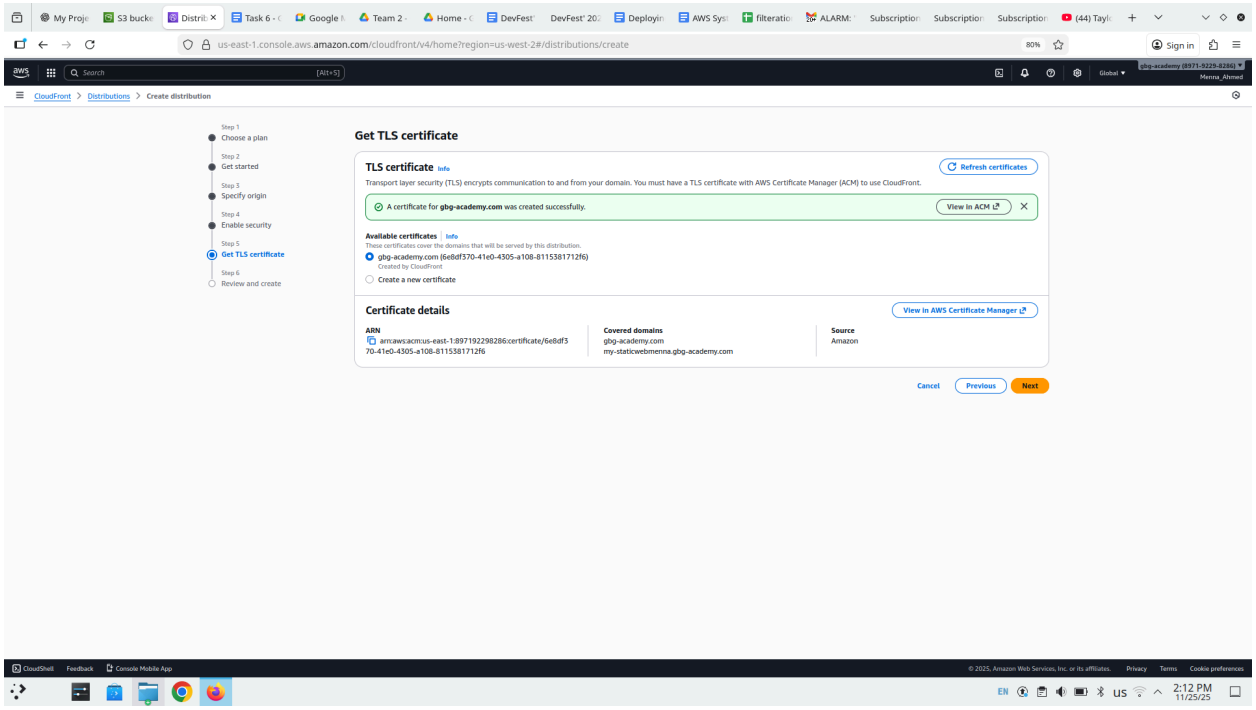
Doesn't needed WAF

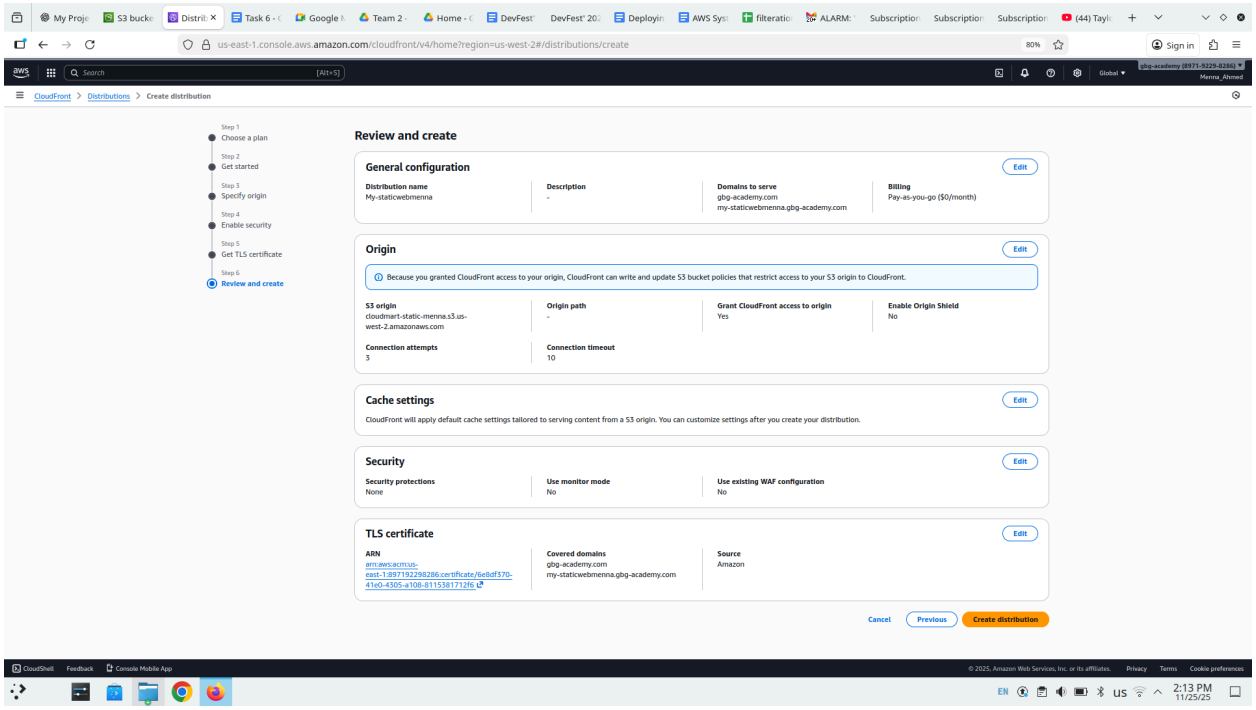


Next

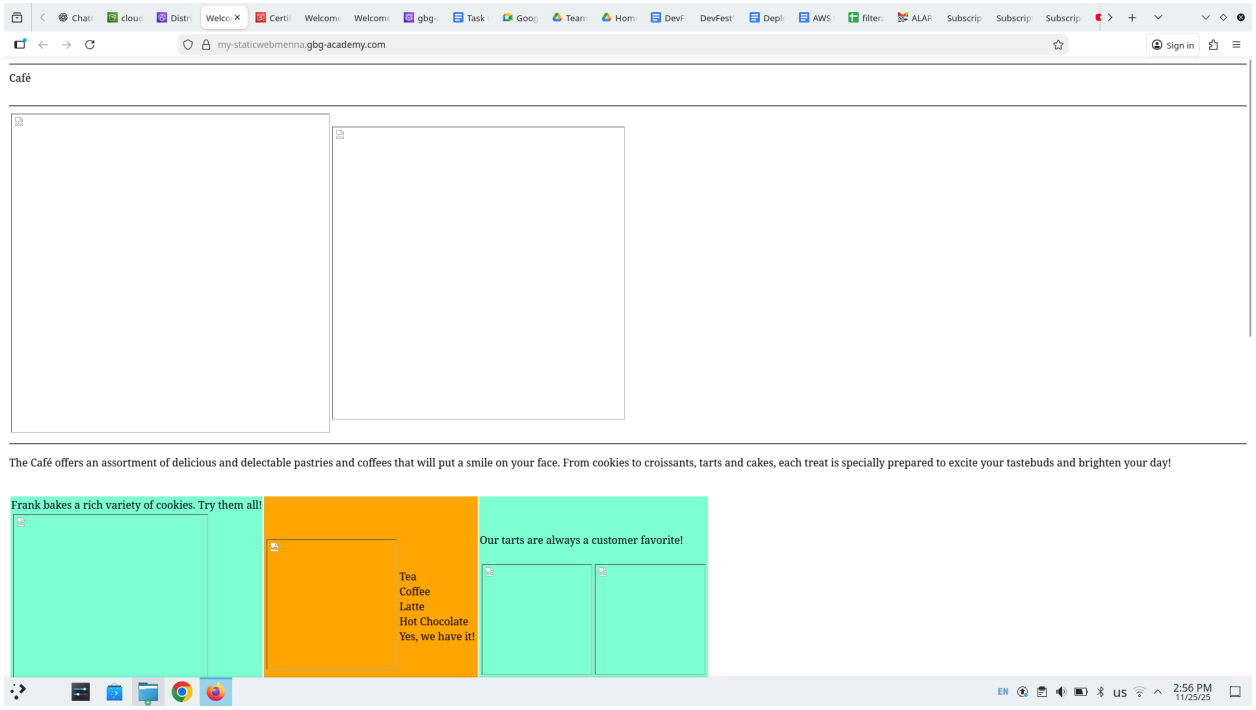


Choose create certificate

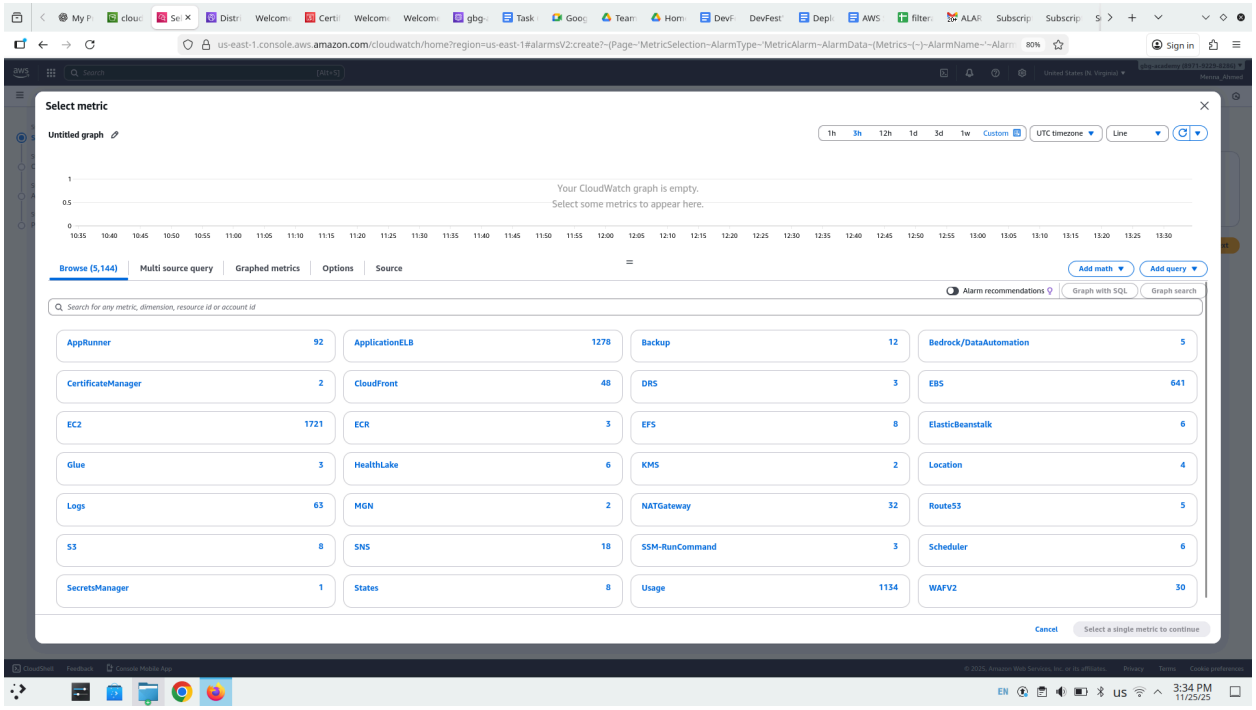




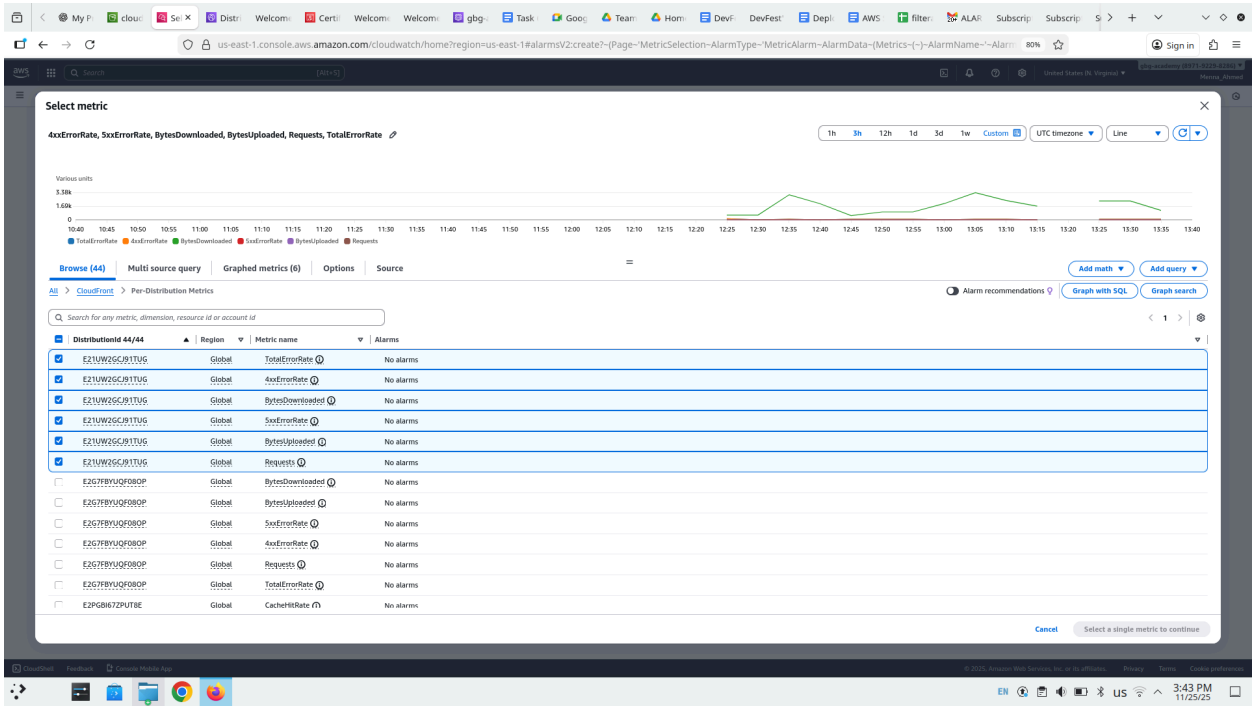
Open link



Create Alarm



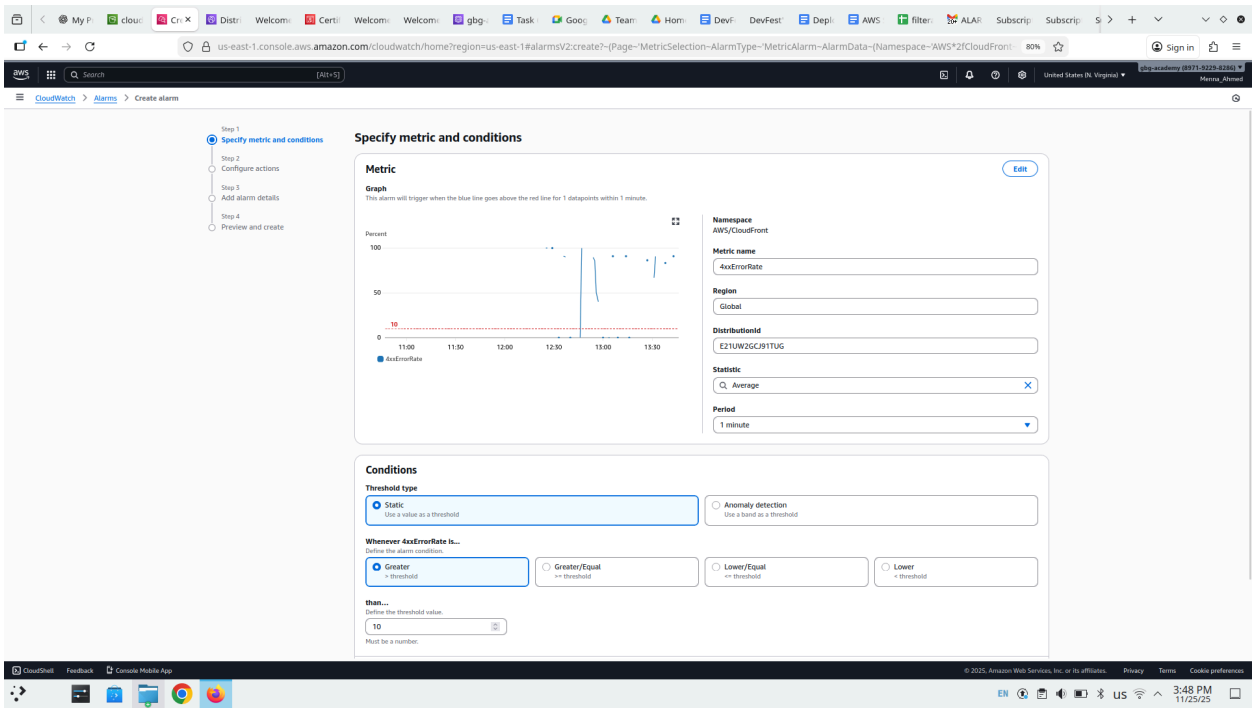
choose cloudfront and you should on north virginia



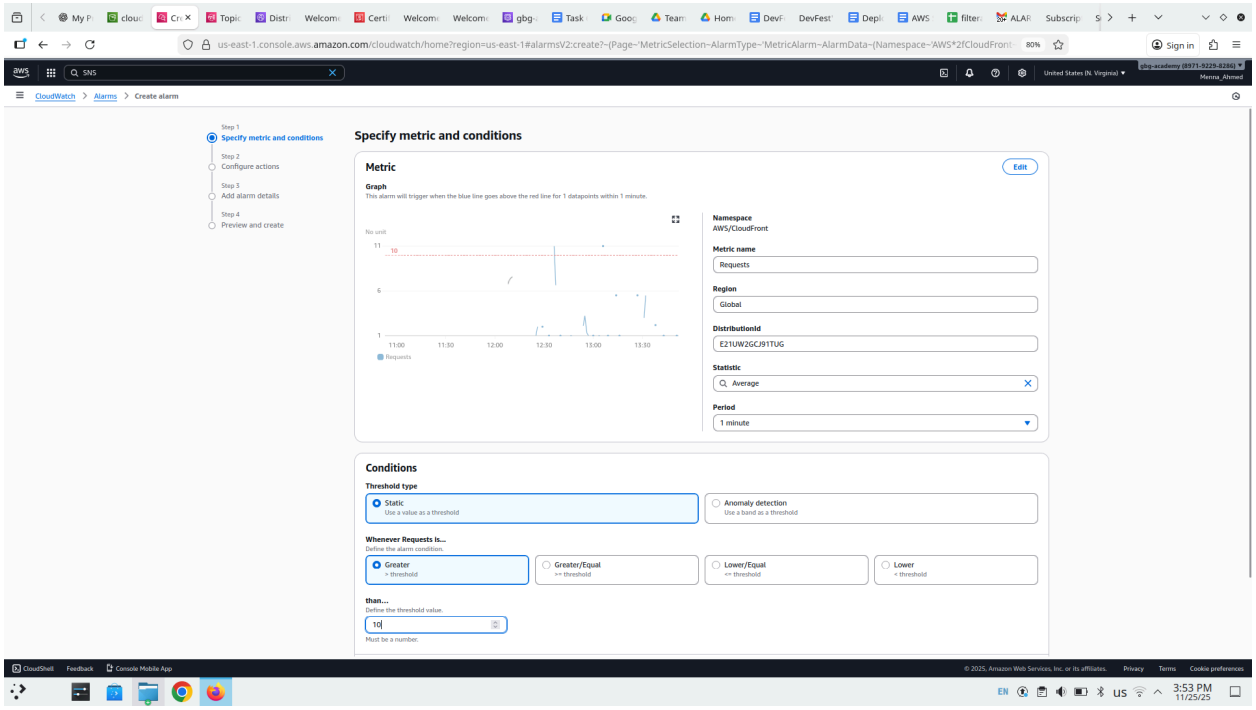
علشان دي اللي فيها الـ metrics الخاصة بلوحة CloudFront بتاعتك.

1- Create Alaem with the metrics

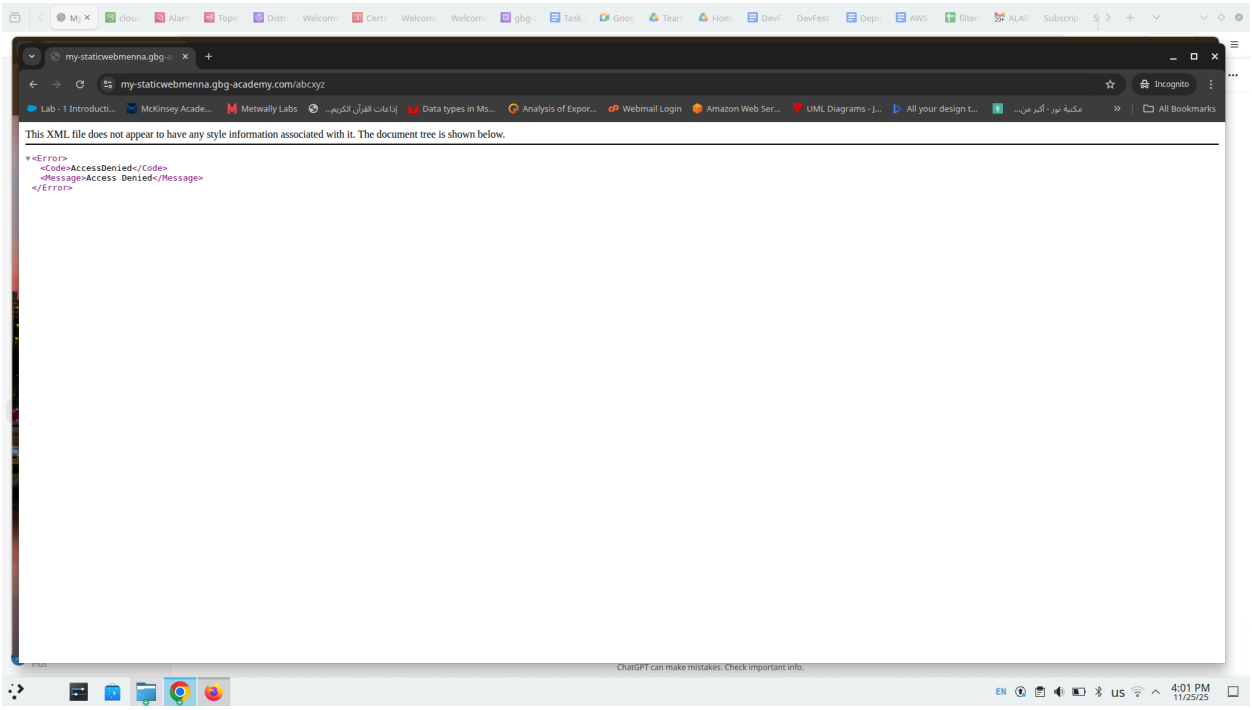
- 4xxErrorRate



- Request Alarm



Make test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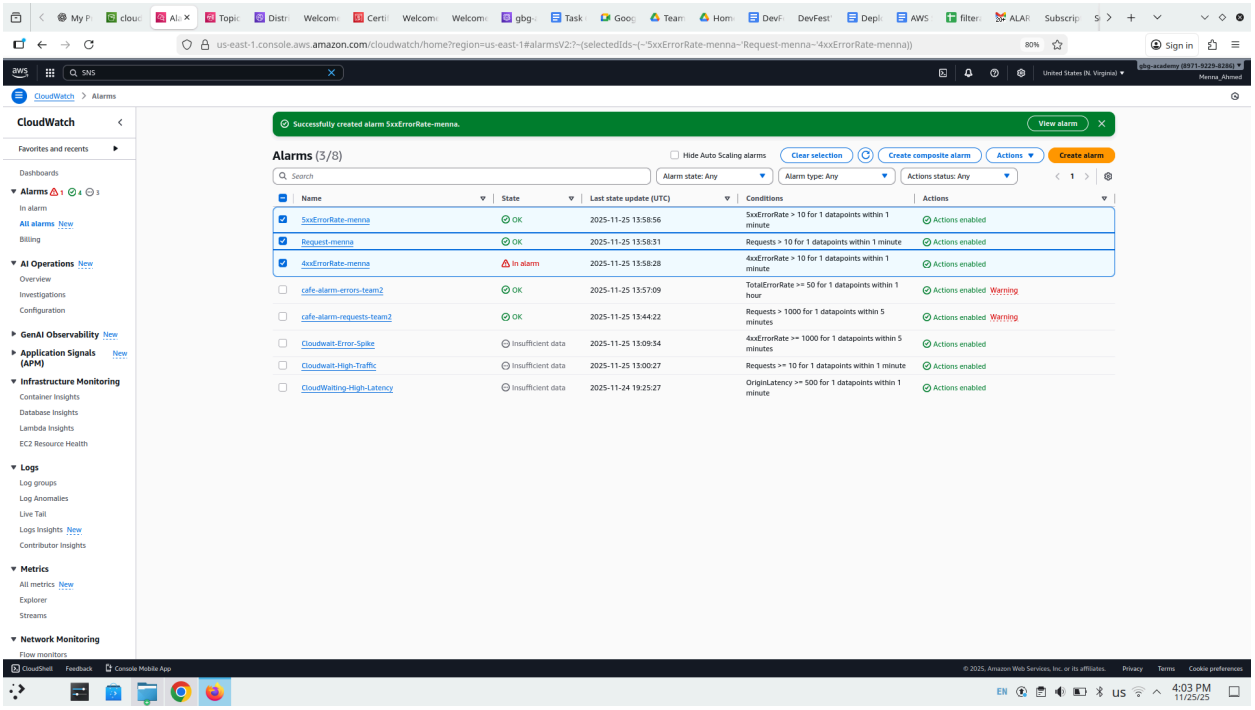
ChatGPT can make mistakes. Check important info.


us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#alarmsV2:?~(selectedIds~(~'5xxErrorRate-menna~'Request-menna~'4xxErrorRate-menna))
CloudWatch > Alarms
Successfully created alarm 5xxErrorRate-menna.
View alarm
Alarms (3/8)
Hide Auto Scaling alarms
Clear selection
Create composite alarm
Actions
Create alarm
Alarm state: Any
Alarm type: Any
Actions status: Any
Name
State
Last state update (UTC)
Conditions
Actions
5xxErrorRate-menna
OK
2025-11-25 13:58:56
5xxErrorRate > 10 for 1 datapoints within 1 minute
Actions enabled
Request-menna
OK
2025-11-25 13:58:31
Requests > 10 for 1 datapoints within 1 minute
Actions enabled
4xxErrorRate-menna
In alarm
2025-11-25 13:58:28
4xxErrorRate > 10 for 1 datapoints within 1 minute
Actions enabled
cafe-alarm-errors-team2
OK
2025-11-25 13:57:09
TotalErrorRate >= 50 for 1 datapoints within 1 hour
Actions enabled
Warning
cafe-alarm-requests-team2
OK
2025-11-25 13:44:22
Requests > 1000 for 1 datapoints within 5 minutes
Actions enabled
Warning
Cloudwait-Error-Spike
Insufficient data
2025-11-25 13:09:34
4xxErrorRate >= 1000 for 1 datapoints within 5 minutes
Actions enabled
Cloudwait-High-Traffic
Insufficient data
2025-11-25 13:00:27
Requests >= 10 for 1 datapoints within 1 minute
Actions enabled
CloudWaiting-High-Latency
Insufficient data
2025-11-24 19:25:27
OriginLatency >= 500 for 1 datapoints within 1 minute
Actions enabled